

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414650
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Tokarz; Steven et al.

Cooking rack

Abstract

A cooking rack according to an example of the present disclosure includes, among other possible things, a top grate extending between a first end portion and a second end portion, a base extending between a first end portion and a second end portion, the base spaced apart from the top grate, a first support supporting the first end portion of the top grate with respect to the first end portion of the base, a second support supporting the second end portion of the top grate with respect to the second end portion of the base, and first and second rails situated between the top grate and the base. A method of cooking and another example cooking rack are also disclosed.

Inventors: Tokarz; Steven (Superior Township, MI), Damico; Mario (Attica, MI), Vaughn; Brennan (Mattawan, MI)

Applicant: Mario's Backyard, LLC (Attica, MI)

Family ID: 1000008819173

Assignee: Mario's Backyard, LLC (Attica, MI)

Appl. No.: 17/690778

Filed: March 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220287503 A1	Sep. 15, 2022

Related U.S. Application Data

us-provisional-application US 63158505 20210309

Publication Classification

Int. Cl.: A47J37/06 (20060101); A47J43/18 (20060101); F24C15/16 (20060101)

U.S. Cl.:

CPC A47J37/0694 (20130101); A47J43/18 (20130101); F24C15/16 (20130101);
A47J37/0664 (20130101)

Field of Classification Search

CPC: A47J (43/18); A47J (37/0694); A47J (37/0664); A47J (37/0878); F24C (15/16)

USPC: 99/426; 99/448; 211/181.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4848217	12/1988	Koziol	99/426	A47J 37/0694
5730046	12/1997	Battaglia et al.	N/A	N/A
5823100	12/1997	Wienhold	248/172	F24C 15/16
6131506	12/1999	Kemper	99/425	A47J 37/0694
6827006	12/2003	D'Amato et al.	N/A	N/A
6945161	12/2004	Battaglia et al.	N/A	N/A
7040219	12/2005	D'Amato et al.	N/A	N/A
2005/0279223	12/2004	McPeak	N/A	N/A
2007/0261569	12/2006	D'Amato et al.	N/A	N/A
2009/0020491	12/2008	Foster	99/450	A47J 37/0786
2011/0113974	12/2010	D'Amato et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2595284	12/2020	GB	A47J 43/18

Primary Examiner: Alexander; Reginald

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/158,505 filed Mar. 9, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Many foods benefit from being elevated or suspended with respect to a heat source during cooking. This allows for improved air circulation around the food for even cooking. Also, minimizing points of contact between the foods and heated surfaces such as a cooking grate of a barbeque reduces the risk of burning or sticking.

(2) Moreover, in certain circumstances it may be advantageous to cook multiple foods simultaneously. However, the foods may have different shapes, sizes, and cooking requirements. This is especially important for chicken drumsticks, ribs, and chicken parts such as wings,

wingdings and breast.

SUMMARY

(3) A cooking rack according to an example of the present disclosure includes, among other possible things, a top grate extending between a first end portion and a second end portion, a base extending between a first end portion and a second end portion, the base spaced apart from the top grate, a first support supporting the first end portion of the top grate with respect to the first end portion of the base, a second support supporting the second end portion of the top grate with respect to the second end portion of the base, and first and second rails situated between the top grate and the base.

(4) A method of cooking food using a cooking rack according to an example of the present disclosure includes, among other possible things, placing a cooking rack on a cooking surface, and arranging food on the cooking rack so that it is suspended with respect to the cooking surface. The cooking rack includes a top grate extending between a first end portion and a second end portion, a base extending between a first end portion and a second end portion, the base spaced apart from the top grate, a first support supporting the first end portion of the top grate with respect to the first end portion of the base, a second support supporting the second end portion of the top grate with respect to the second end portion of the base, first and second rails situated between the top grate and the base.

(5) A cooking rack according to an example of the present disclosure includes, among other possible things, a top grate extending between a first end portion and a second end portion, a base extending between a first end portion and a second end portion, the base spaced apart from the top grate, a first support supporting the first end portion of the top grate with respect to the first end portion of the base, a second support supporting the second end portion of the top grate with respect to the second end portion of the base, and a side hanger extending outwardly from the top grate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example a cooking rack.

(2) FIG. 2 shows a perspective view of the cooking rack of FIG. 1.

(3) FIG. 3 shows another perspective view of the cooking rack of FIG. 1.

(4) FIG. 4 shows a side view of the cooking rack of FIG. 1.

(5) FIG. 5 shows another view of the cooking rack of FIG. 1.

DETAILED DESCRIPTION

(6) The present disclosure relates to a cooking rack that can be used on a hot environment such as an open fire, oven, barbeque, or smoker to hold foods in an elevated or suspended position with respect to a heat source.

(7) FIG. 1 illustrates an example cooking rack **10**. The rack **10** includes a base **12** and a top grate **14** spaced apart from the base **12**. As shown, the base **12** is supported on the top grate **14** via supports **16**.

(8) Referring to FIG. 2 with continued reference to FIG. 1, the base **12** extends between a first end portion **18** and a second end portion **20**. Each of the first and second end portions **18**, **20** can include feet **22**. In one example, the feet **22** extend perpendicularly outward with respect to the supports **16** such that the feet **22** have a width **W**. Similarly, the top grate **14** extends between a first end portion **24** and a second end portion **26**. Each of the first and second end portions **24**, **26** can include one or more handles **28**. In one example, the handles **28** also extend perpendicularly outward with respect to the supports **16**. In some examples, the placement and/or geometry of the handles **28** corresponds to the placement of the feet **22**. The handles **28** allow a user to readily place

the rack **10** where desired and/or move the rack **10**.

(9) In one example, the cooking rack **10** can be placed directly onto a cooking surface, such as a grate of a barbeque or directly onto an oven rack. In these examples, the width **W** of the feet **22** is greater than the spacing of the grate of the barbeque or oven rack so that the feet **22** stabilize the rack **10** and allow it to sit flat on the cooking surface. In another example, the cooking rack **10** can be placed onto a baking sheet or pan. The same can occur in an open fire, smoker, or similar heat source.

(10) In one example, as shown in FIG. 3, the supports **16** can include a first support **16A**, a second support **16B**, a third support **16C**, and a fourth support **16D**. In this example, the first and second supports **16A/16B** are configured to connect the first end portion **18** of the base **12** to the first end portion **24** of the top grate **14**. In addition, the third and fourth supports **16C/16D** are configured to connect the second end portion **20** of the base **12** to the second end portion **26** of the top grate **14**. In this example, the base **12** and top grate **14** have a rectangular shape defined by a length and a width, and four corners. Accordingly, one of the supports **16A/16B/16C/16D** is positioned at each of the four corners. The supports **16** space the base **12** apart from the top grate **14**.

(11) Referring back to FIGS. 1-2, the top grate **14** includes rods **30** that extend across the top grate **14**. Between adjacent rods **30** are gaps **32**. In some examples, the gaps **32** are smaller than the width **W** of the feet **22**. The rods **30** can extend across the width of the cooking rack **10** where the top grate **14**/base **16** have a generally rectangular shape. The diameter of the rods **30** can be balanced with the number of rods **30**. The rods **30** are configured to support a wide variety of foods for cooking, including but not limited to sausages, seafood such as salmon filets, steaks, etc. Because the top grate **14** is elevated from the cooking surface, it allows for more even circulation of hot air around the food, thus encouraging even cooking and reducing the risk of the food burning or sticking to the cooking surface.

(12) In addition to the top grate **14**, the rack **10** includes one or more optional components for holding food elevated/suspended from the cooking surface.

(13) Referring to FIGS. 4 and 5, in one example, the optional component is at least one set of rails **34/36** between the base **12** and the top grate **14**. In the example of FIG. 4, there are two sets of rails **34/36**, but in other examples there could be one set of rails or more than two sets of rails. In this example, the first set of rails **34** includes a first outer rail **38** and a first inner rail **40**. Similarly, the second set of rails **36** includes a second inner rail **42** and a second outer rail **44**. The sets of rails **34/36** extend along the length of the top grate **14** on opposite sides of the top grate **14** where the top grate **14**/base **16** have a generally rectangular shape. In a particular example, the inner rails **40/42** in each set of rails **34/36** is connected to the top grate **14** by connectors **46**. The outer rails **38/44** in each set of rails **34/36** extend between supports **16**.

(14) The rails in each set of rails **34/36** are spaced apart by a distance **D**. The distance **D** is selected to hold the thinner end of a chicken drumstick between the rails **34/36**. In some examples, the distance **D** is about 1 inch. One or more chicken drumsticks can be slid in between the rails **34/36**, and thus the chicken drumsticks can be suspended from the sets of rails **34/36** over the cooking surface, with the thicker end closer to the cooking surface and the thinner end further from the cooking surface. As discussed above, suspending the chicken drumsticks in this way allows for even cooking and reduces the risk of the chicken drumsticks from burning or sticking to the cooking surface. Moreover, juices released from the chicken drumsticks during cooking are pulled down the length of the drumsticks by gravity, keeping the chicken drumsticks moist.

(15) Referring back to FIG. 3, another example component is one or more brackets **48**. The brackets **48** can extend from individual rods **30** of the top grate **14**, and in some examples can be integral with individual rods **30** of the top grate **14**. In one example, the brackets **48** include large brackets **50** and small brackets **52**. The brackets **50** include a downward extension **50A**, which extends down from the top grate **14** towards the base **12**, and an outward extension **50B** which extends outward from the downward extension **50A** to provide an “L” shape. Similarly, the small

brackets **52** include a downward extension **52A**, which extends down from the top grate **14** towards the base **12**, and an outward extension **52B** which extends outward from the downward extension **52A** to provide an “L” shape.

(16) In some examples, large brackets **50** and small brackets **52** may be provided in an alternating or another arrangement. The small brackets **52** may have smaller downward extensions **52A**, outward extensions **52B**, or both as compared to the large brackets **50**. In some examples, the outward extensions **50B/52B** extends outward from the downward extensions **50B/52B** at an angle equal to or less than about 90 degrees.

(17) As used herein, the term “about” has the typical meaning in the art, however in a particular example “about” can mean deviations of up to 30% of the values described herein.

(18) The brackets **48** can support foods that have at least one small dimension, such as sausages, chicken wings, or vegetables. Foods can either rest on top of the outward extension(s) **50B/52B** of the bracket(s) **50B/52B** or can be pierced onto the outward extension(s) **50B/52B** of the bracket(s) **50/52**. In one example, ends of the outward extensions **50B/52B** have a diameter that is less than the diameter of the rods **30**. As a result, the ends of the outward extensions **50B/52B** are sharp enough to pierce food. As with the rails **34/36**, the brackets **48** allow food to be suspended from the cooking surface, which allows for more even air cooking due to improved air circulation around the foods.

(19) With further reference to FIG. 5, another example component is side hangers **54**. In one example, the side hangers **54** can extend outward from the top grate **14**. More particularly, the side hanger **54** can hang from the outer rails **38/44** of the sets of rails **34/36** or from the supports **16**. The side hangers **54** have a corrugated structure that defines first and second alternating sets of openings **56/58**. The sets of openings **56/58** can have the same or different geometries. In one example, a first set of openings **56** has a triangular geometry and the second set of openings **58** has a trapezoidal geometry with squared off ends. In the example of FIG. 5, the side hangers **54** are formed of a single continuous piece. For instance, the single piece side hangers **54** includes a unitary structure formed into the corrugated shape. In other examples, the side hangers **54** are formed of multiple separate pieces that together define the corrugated structure.

(20) In some examples, the side hangers **54** may hang such that each one of the side hangers **54** are oriented approximately perpendicular to the top grate **14**. In some examples, the side hangers **54** may be angled with respect to the top grate **14** such that each one of the side hangers **54** is angled away from a centerline C of the rack **10**. The openings **56/58** are configured to receive small foods such as chicken wings as well as slabs of meat or ribs and support the foods away from the cooking surface. As with the rails **34/36** and brackets **48**, the side hanger **26** allows for more even air cooking due to improved air circulation around the foods.

(21) The various components described above may be removable from the rack **10** for easy cleaning and storage, in some examples.

(22) The cooking rack **10** can be made from stainless steel or another suitable material. In some examples, the material prevents or reduces sticking of foods to the rack **10**. Non-stick or stick-resistant materials suitable for cooking are well known in the art.

(23) The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from this disclosure. The scope of legal protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A cooking rack, comprising: a top grate extending between a first end portion and a second end portion; a base extending between a first end portion and a second end portion, the base spaced apart from the top grate; a first support supporting the first end portion of the top grate with respect

to the first end portion of the base; a second support supporting the second end portion of the top grate with respect to the second end portion of the base; first and second rails situated between the top grate and the base, and a plurality of rods extending across the top grate, wherein at least one of the plurality of rods includes a bracket, and the bracket includes a downward extension that extends down from the top grate towards the base, and an outward extension that extends outwards from the downward extension.

2. The cooking rack of claim 1, wherein the bracket includes a first bracket and a second bracket, wherein the downward extension of the second bracket is smaller than the downward extension of the first bracket or the outward extension of the second bracket is smaller than the outward extension of the first bracket.

3. The cooking rack of claim 1, wherein the bracket is configured to receive food on the outward extension.

4. The cooking rack of claim 1, wherein the outward extension is configured to pierce through food.

5. The cooking rack of claim 1, wherein each of the first and second end portions of the top grate includes a handle.

6. The cooking rack of claim 1, wherein each of the first and second end portions of the base includes a foot.

7. The cooking rack of claim 1, wherein the first and second rails are spaced apart by a distance D .

8. The cooking rack of claim 7, wherein a space between the first and second rails is configured to receive food such that the food is held between the first and second rails and suspended below the top grate.

9. The cooking rack of claim 1, further comprising a side hanger extending from at least one of the first and second supports or at least one of the first and second rails.

10. The cooking rack of claim 9, wherein the side hanger includes a corrugated structure that defines first and second sets of alternating openings.

11. The cooking rack of claim 1, wherein the first and second rails are on a first side of the cooking rack, and further comprising third and fourth rails situated between the top grate and the base on a second side of the cooking rack opposite the first side.

12. The cooking rack of claim 11, wherein each of the top grate and the base have a rectangular shape defined by a length and a width, and four corners, and further comprising third and fourth supports such that one of the first, second, third, and fourth supports is positioned at each corner.

13. A cooking rack, comprising: a top grate extending between a first end portion and a second end portion; a base extending between a first end portion and a second end portion, the base spaced apart from the top grate; a first support supporting the first end portion of the top grate with respect to the first end portion of the base; a second support supporting the second end portion of the top grate with respect to the second end portion of the base; a side hanger extending outwardly from the top grate; first and second rails situated between the top grate and the base; and a plurality of rods extending across the top grate, wherein at least one of the plurality of rods includes a bracket, and the bracket includes a downward extension that extends down from the top grate towards the base, and an outward extension that extends outwards from the downward extension.

14. The cooking rack of claim 13, wherein the side hanger includes a corrugated structure that defines first and second sets of alternating openings, the first set of openings have a triangular geometry and the second set of opening have a trapezoidal geometry with squared-off ends, and the first set of openings is configured to receive food therein.

15. A cooking rack, comprising: a top grate extending between a first end portion and a second end portion; a base extending between a first end portion and a second end portion, the base spaced apart from the top grate; a first support supporting the first end portion of the top grate with respect to the first end portion of the base; a second support supporting the second end portion of the top grate with respect to the second end portion of the base; first and second rails situated between the

top grate and the base; and a side hanger extending from at least one of the first and second supports or at least one of the first and second rails.

16. The cooking rack of claim 15, wherein the side hanger includes a corrugated structure that defines first and second sets of alternating openings.
